NOTES ON PROOF ROUTE CHOICES IN THE WEIL CONJECTURES BLUEPRINT

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Abstract. We explain the rationale behind the choice of constructions and proofs in our blueprint for the Weil conjectures. Our approach favors ordinary categories, the reuse of existing foundations, and a small collection of widely applicable tools.

This note records the rationale behind our proof-route choices in the Weil conjectures. We first state the general principles that guide the proof as a whole (§1), then discuss the choices made for individual topics (§2). The route itself uses modern methods for a good part of the argument, some of which have not been written down in this setting before.

1. Guiding principles

Ordinary categories. We work only with 1-categories. Using higher category theory, such as 2-categories or ∞-categories, would make the proof depend on an unnecessarily large input; formalizing it would also considerably enlarge the project without being necessary for the results we use.

Avoiding local acyclicity and vanishing cycles. We avoid the general formalism of local acyclicity and of nearby and vanishing cycles [SGA7II, Exp. XIII]. Besides its technical geometric inputs, this formalism introduces delicate functoriality and equivariance issues. In the derived 1-category, taking a cone is not in general functorial, so defining vanishing cycles as a cone does not by itself provide their functoriality; higher-categorical enhancements express this functoriality and its coherences more naturally. Avoiding this formalism therefore keeps the proof within our chosen foundations and reduces the work needed to manage Galois actions and specialization.

Weil II and ℓ-adic coefficients. We choose the route via weight theory in *Weil II* [D2] over the route via Lefschetz pencils in *Weil I* [D1]: the result of *Weil II* is more general than that of *Weil I*; weight theory and the Fourier transform tend to be used more broadly in the literature than Lefschetz-pencil theory; and the improved Fourier-transform proof adopted here avoids vanishing cycles (§2.6), whereas the Lefschetz-pencil proof relies on them.

For coefficients we choose ℓ-adic, specifically pro-étale, cohomology. Pro-étale cohomology is already available in Mathlib, and the `proetale` repository [JM] formalizes the comparison theorems with étale cohomology, so this material can be reused rather than developed from scratch. By contrast, the p-adic alternative relevant here is rigid cohomology, which would require developing a separate foundation, including rigid analytic spaces.

2. Choices for individual topics

The choices below are grouped by topic; each records what we chose and why.

2.1. Constructible derived categories

Pro-étale rather than classical adic systems. The pro-étale topology of Bhatt–Scholze [BS, §§3–6] gives a cleaner treatment of adic coefficients. Classically, coefficients are encoded by systems of étale sheaves; in the pro-étale setting, one has repleteness and actual coefficient sheaves $\mathcal{O}_{E,X}$ and E_X for an arbitrary algebraic extension $E/\mathbb{Q}_\ell$. This makes the constructible categories more natural to formulate. In the noetherian setting it also simplifies the definitions and provides a uniform foundation for the six functors needed later [BS, §6.8].

The condensed approach. We also use the perspective of Hemo–Richarz–Scholbach [HRS, Def. 3.3, §7], which treats lisse and constructible objects with finite, $\mathcal{O}_E$- or E-coefficients uniformly on arbitrary schemes. Although their paper is written using ∞-categories, we need only selected results that can be stated and proved using ordinary categories and derived 1-categories. We adopt these results for their uniformity and generality without taking on the full higher-categorical framework of the paper.

2.2. Proper and smooth base change

We use the Stacks Project proofs of proper and smooth base change [Stacks, Tags `095T`, `0EYU`]. The proper-base-change argument remains close to SGA 4, while the direct smooth-base-change argument avoids the local-acyclicity formalism used in the classical organization [SGA4, Exp. XVI]. Their use of curve cohomology fits well with the Jacobian theory that is also needed later in this project.

2.3. Smooth purity

For purity of smooth pairs, we choose the method of deformation to the normal cone. It uses normal bundles, vector bundles, and $\mathbb{A}^1$-invariance, which are useful beyond this particular theorem. This is more rewarding for the reader, since the same construction is used in intersection theory [Ful, Chap. 5] and homotopy purity [MV, §3.2].

2.4. The trace formula

Lefschetz–Verdier rather than curve correspondences. We use the Lefschetz–Verdier pairing of SGA 5 [SGA5, Exp. III]. The Nielsen–Wecken type route in SGA $4\frac{1}{2}$ [SGA$4\frac{1}{2}$, *Rapport sur la formule des traces*], also followed by the Stacks Project [Stacks, Tags `03U1`, `03UF`], uses Weil's fixed-point theorem for curves [Wei]. Taking that route would require developing Weil's classical algebraic geometry and classical, non-derived intersection theory. The Lefschetz–Verdier pairing instead reuses our six-functor formalism and is useful for later applications.

The Künneth formula. For the ordinary, non-compact-support Künneth formula needed in the Künneth–Verdier isomorphism, we choose the direct Stacks Project proof [Stacks, Tag `0F1P`] to avoid the general acyclicity machinery. For the pairing itself we follow SGA 5 rather than the relative formulation of Lu–Zheng [LZ]: the latter's 2-categorical framework and broader relative setting would add more work beyond our application over a finite field.

2.5. Weil sheaves

Rank-one geometric monodromy. For smooth geometrically connected curves, we use the proof based on the ℓ-adic logarithm and Jacobians, which are already available for computing the cohomology of curves in the proof of the base change theorems. This replaces the proof in [D2, §1.3], which uses class field theory, and thereby avoids developing it.

Monodromy and determinant weights. We use variants of the arguments in [D2, §1.3] that isolate the facts about reductive groups needed for the determinant weight results. These require some basic results on reductive groups, instead of the classification of reductive groups.

The twist lemma. We use an elementary proof of the twist reduction for irreducible lisse Weil sheaves, rather than deriving it from the geometric monodromy results as in [D2, Prop. 1.3.14]. This reduces its prerequisites of the trace formula for Weil sheaves.

2.6. The Deligne–Laumon estimate

We use a further refinement of the Fourier–Deligne approach, taking Kiehl–Weissauer's treatment [KW, Chap. I] as a reference. This version is not Laumon's original argument [Lau], but a refinement that avoids vanishing cycles. Compared with Deligne's original proof [D2, §3] and Laumon's original development, it has the advantage of a smaller set of prerequisites compatible with our 1-categorical foundations.

We restrict perverse sheaf theory to the perverse t-structure and middle extension on curves. This is enough for the chosen estimate and avoids the cost of developing the general theory in all dimensions. The resulting weight theorem retains the generality that motivated our choice of the Weil II route.

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